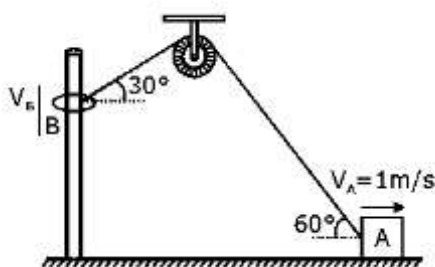


PHYSICS

1. A block of mass m at the end of a string is whirled round in a vertical circle of radius R . The critical speed of the block at the top of its swing below which the string would slacken before the block reaches the top is

(1) Rg (2) $2Rg$ (3) R/g (4) $\sqrt{Rg}$

2. Find velocity of ring B (V_B) at the instant shown. The string is taut and inextensible:

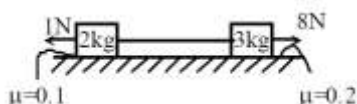


(1) $\frac{1}{2}$ m/s (2) $\frac{\sqrt{3}}{4}$ m/s
(3) $\frac{1}{4}$ m/s (4) 1 m/s

3. A uniform rope of length ℓ lies on a table. If the coefficient of friction is μ , then the maximum length ℓ_1 of the part of this rope which can overhang from the edge of the table without sliding down is

(1) $\frac{1}{\mu}$ (2) $\frac{1}{\mu + 1}$
(3) $\frac{\mu\ell}{1 + \mu}$ (4) $\frac{\mu\ell}{1 - \mu}$

4. In the shown arrangement if f_1 , f_2 and T be the frictional forces on 2 kg block, 3 kg block and tension in the string respectively, then their values are:



(1) 2 N, 6 N, 3.2 N
(2) 2 N, 6 N, 0 N
(3) 1 N, 6 N, 2 N
(4) Data insufficient to calculate the required values

5. A body is sliding down an inclined plane having coefficient of friction 0.5. If the normal reaction is twice that of the resultant downward force along the incline, the angle between the inclined plane and the horizontal is

(1) 15° (2) 30°
(3) 45° (4) 60°

6. A block of mass 1 kg slides down on a rough inclined plane of inclination 60° starting from its top. If the coefficient of kinetic friction is 0.5 and length of the plane is 1 m, then work done against friction is (Take $g = 9.8 \text{ m/s}^2$)

(1) 9.82 J (2) 4.94 J
(3) 2.45 J (4) 1.96 J

7. A uniform chain of length 2m is kept on a table such that a length of 60cm hangs freely from the edge of the table. The total mass of the chain is 4kg. What is the work done in pulling the entire chain on the table

(1) 7.2 J (2) 3.6 J
(3) 120 J (4) 1200 J

8. A spring when stretched by 2 mm its potential energy becomes 4 J. If it is stretched by 10 mm, its potential energy is equal to

(1) 4 J (2) 54 J
(3) 415 J (4) None

9. Water in a bucket is whirled in a vertical circle with a string attached to it. The water does not fall down even when the bucket is inverted at the top of its path. We conclude that:

(1) $mg = \frac{mv^2}{R}$ (2) $mg > \frac{mv^2}{R}$
(3) $mg < \frac{mv^2}{R}$ (4) None of these

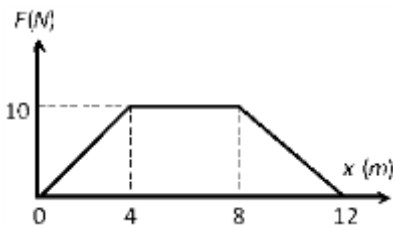
10. A position dependent force $F = 7 - 2x + 3x^2$ newton acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5\text{m}$. The work done in joules is

(1) 70 (2) 270
(3) 35 (4) 135

11. A particle rests on the top of a hemisphere of radius R . Find the smallest horizontal velocity that must be imparted to the particle if it is to leave the hemisphere without sliding down it-

(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$
(3) $\sqrt{3gR}$ (4) $\sqrt{5gR}$

12. A particle of mass 0.1 kg is subjected to a force which varies with distance as shown in fig. If it starts its journey from rest at $x = 0$, its velocity at $x = 12 \text{ m}$ is -

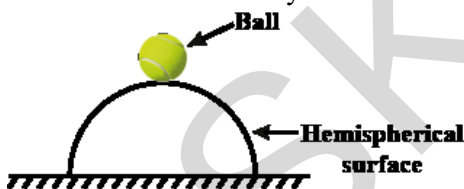


(1) 0 m/s (2) $20\sqrt{2} \text{ m/s}$
(3) $20\sqrt{3} \text{ m/s}$ (4) 40 m/s

13. If the heart pushes 1 cc of blood in one second under pressure 20000 N/m^2 the power of heart is -

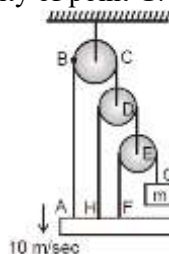
(1) 0.02 W (2) 400 W
(3) $5 \times 10^{-10} \text{ W}$ (4) 0.2 W

14. A ball is placed on the top hemispherical bowl as shown in the figure. Which type of equilibrium can be associated with the body?



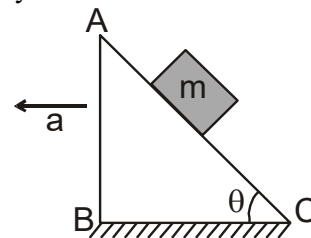
(1) Neutral equilibrium
(2) Stable equilibrium
(3) Unstable equilibrium
(4) Insufficient equilibrium

15. Find the velocity of point G.



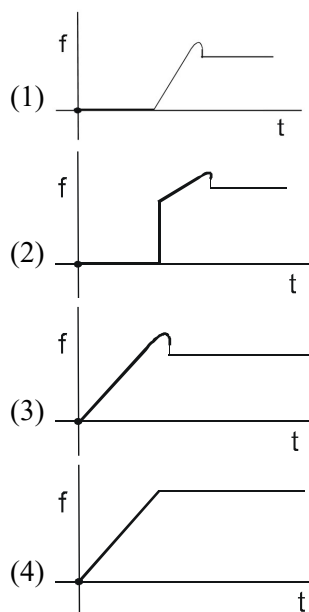
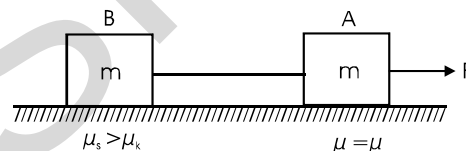
(1) 50 m/s (2) 7 m/s
(3) 70 m/s (4) None

16. A block of mass m resting on a wedge of angle θ as shown in the figure. The wedge is given an acceleration a towards left. What is the minimum value of a due to external agent so that the mass m falls freely?



(1) g (2) $g \cos \theta$
(3) $g \cot \theta$ (4) $g \tan \theta$

17. A force $F = t$ is applied to block A as shown in figure. The force is applied at $t = 0$ seconds when the system was at rest and string is just straight without tension. Which of the following graphs gives the friction force between B and horizontal surface as a function of time 't'.



18. A car weight 1400 kg is moving at a speed of 54 kmh^{-1} up a hill. When the motor stops, it is just able to reach the destination which is at a height of 10 m above the point. Then the work done against friction (negative of the work done by the friction) is (Take, $g = 10 \text{ ms}^{-2}$)

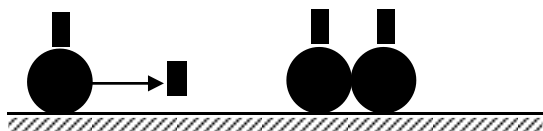
(1) 10 kJ (2) 15 kJ
(3) 17.5 kJ (4) 25 kJ

19. A ball is released from the top of a tower. The ratio of work done by force of gravity in 1st second, 2nd second and 3rd second of the motion of ball is
 (1) 1 : 2 : 3 (2) 1 : 4 : 16
 (3) 1 : 3 : 5 (4) 1 : 9 : 25

20. When a man increases his speed by 2 ms^{-1} , he finds that his kinetic energy is doubled then the original speed of the man is

- (1) $2(\sqrt{2}-1) \text{ ms}^{-1}$ (2) $2(\sqrt{2}+1) \text{ ms}^{-1}$
 (3) 4.5 ms^{-1} (4) None of these

21. Two identical balls marked 2 and 3, in contact with each other and at rest on a horizontal frictionless table, are hit head on by another identical ball marked 1, which is moving initially with a speed v as shown in figure. What is observed, if the collision is elastic?



- (1) Ball 1 comes to rest and balls 2 and 3 roll out with speed $\frac{v}{2}$ each
 (2) Balls 1 and 2 come to rest and ball 3 rolls out with speed v
 (3) Balls 1, 2 and 3 roll out with speed $\frac{v}{3}$ each
 (4) Balls 1, 2 and 3 come to rest

22. A neutron moving at a speed ' v ' undergoes a head-on elastic collision with a nucleus of mass number A at rest. The ratio of the kinetic energies of the neutron after and before collision is:

- (1) $\left(\frac{A-1}{A+1}\right)^2$ (2) $\left(\frac{A+1}{A-1}\right)^2$
 (3) $\left(\frac{A}{A+1}\right)^2$ (4) $\left(\frac{A}{A-1}\right)^2$

23. A shell of mass m is at rest initially. It explodes into three fragments having masses in the ratio 2 : 2 : 1. The fragments having equal masses fly off along mutually perpendicular directions with speed v . What will be the speed of the third (lighter) fragment?

- (1) v (2) $\sqrt{2}v$
 (3) $2\sqrt{2}v$ (4) $3\sqrt{2}v$

24. A ball of mass m moving horizontally at a speed v collides with the bob of a simple pendulum at rest. The mass of the bob is also m . If the collision is perfectly inelastic and both balls sticks, the height to which the two balls rise after the collision will be given by:

- (1) $\frac{v^2}{g}$ (2) $\frac{v^2}{2g}$ (3) $\frac{v^2}{4g}$ (4) $\frac{v^2}{8g}$

25. A metal ball does not rebound when struck on a wall, whereas rubber ball of same mass when thrown with the same velocity on the wall rebounds. From this it is inferred that -

- (1) Change in momentum is same in both
 (2) Change in momentum in rubber ball is more
 (3) Change in momentum in metal ball is more
 (4) Initial momentum of metal ball is more than that of rubber ball

26. In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles after collision, is.

- (1) $\frac{v_0}{4}$ (2) $\sqrt{2}v_0$ (3) $\frac{v_0}{2}$ (4) $\frac{v_0}{\sqrt{2}}$

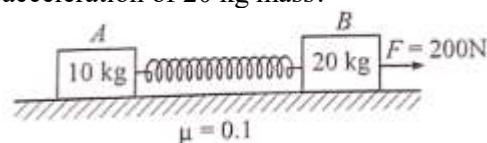
27. A molecule in a gas container hits a horizontal wall with speed 200 ms^{-1} at an angle 30° with the normal and rebounds with the same speed. Which of the following option is correct?

- (1) Momentum is not conserved.
 (2) Elastic collision occurs here.
 (3) Inelastic collision occurs here.
 (4) Both (1) and (2)

28. Essential characteristic of equilibrium is:

- (1) Momentum equal zero
 (2) Acceleration equals zero
 (3) K.E. equals zero
 (4) Velocity equals zero

29. Two blocks A and B, attached to each other by a massless spring, are kept on a rough horizontal surface ($\mu = 0.1$) and pulled by a force $F = 200 \text{ N}$ as shown in the figure. If at some instant, the 10 kg mass has acceleration of 12 m/s^2 then what is the acceleration of 20 kg mass?



- (1) 2.5 m/s^2 (2) 4.0 m/s^2
 (3) 3.6 m/s^2 (4) 1.2 m/s^2

30. Twelve one-rupee coins are put on top of each other on a table, where mass of each coin is 20 g. The reaction of the 7th coin (counted from bottom) on the 8th coin is (take, $g = 10 \text{ m/s}^2$).

(1) 2 N (2) 1 N
(3) 3 N (4) 4 N

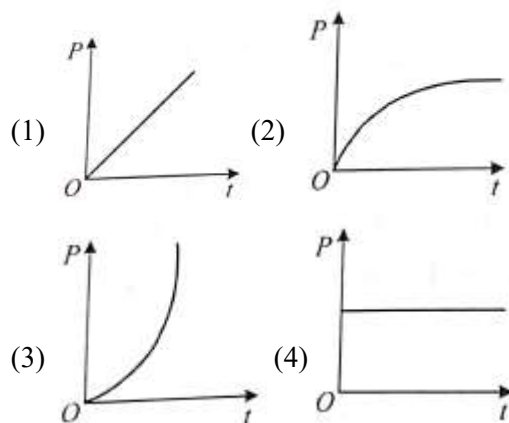
31. If a box is lying in the compartment of an accelerating train and box is stationary relative to the train. What force cause the acceleration of the box?

(1) Frictional force in the direction of train
(2) Frictional force in the opposite direction of train
(3) Force applied by air
(4) None of the above

32. A particle in a certain conservative force field has a potential energy given by $U = \frac{10xy}{z}$. The force exerted on it is

(1) $\left(\frac{10y}{z}\right)\hat{i} + \left(\frac{10x}{z}\right)\hat{j} + \left(\frac{10xy}{z^2}\right)\hat{k}$
(2) $-\left(\frac{10y}{z}\right)\hat{i} - \left(\frac{10x}{z}\right)\hat{j} + \left(\frac{10xy}{z^2}\right)\hat{k}$
(3) $-\left(\frac{10y}{z}\right)\hat{i} - \left(\frac{10x}{z}\right)\hat{j} - \left(\frac{10xy}{z^2}\right)\hat{k}$
(4) $\left(\frac{10y}{z}\right)\hat{i} + \left(\frac{10x}{z}\right)\hat{j} - \left(\frac{10xy}{z^2}\right)\hat{k}$

33. A motor drives a body along a straight line with a constant force. The power P developed by the motor must vary with time t as:



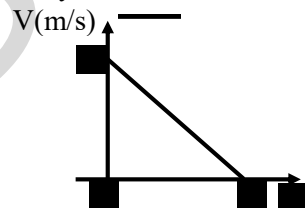
34. A man pulls a bucket of water from a depth of h from a well. If the mass of the rope and that of bucket full of water are m and M respectively, the work done by the man is:

(1) $(M+m)gh$ (2) $\left(M + \frac{m}{2}\right)gh$
(3) $\left(\frac{M+m}{2}\right)gh$ (4) $\left(\frac{M}{2} + m\right)gh$

35. A particle of mass 0.01 kg travels along a space curve with velocity given by $(4\hat{i} + 16\hat{k}) \text{ m/s}$. After some time, its velocity becomes $(8\hat{i} + 20\hat{j}) \text{ m/s}$ due to the action of a conservative force. The work done on the particle during this interval of time is:

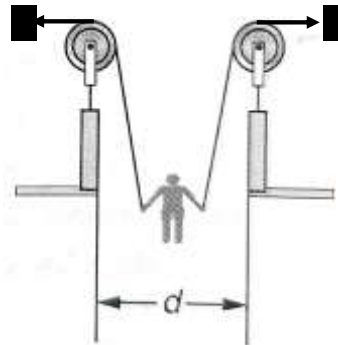
(1) 0.32 J (2) 6.9 J
(3) 9.6 J (4) 0.96 J

36. Velocity-time graph of a particle of mass 2 kg moving in a straight line is as shown in figure. Work done by all the forces on the particle is:



(1) 400 J (2) -400 J
(3) -200 J (4) 200 J

37. A man of mass m has fallen into a ditch of width d . Two of his friends are slowly pulling him out using a light rope and two fixed pulleys as shown in figure. Both the friends exert force of equal magnitudes F . When the man is at a depth h , the value of F is:



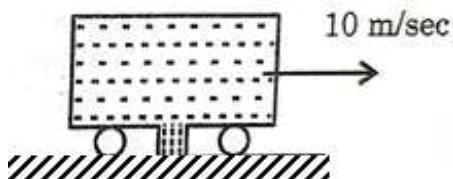
(1) $\frac{mg}{4h} \sqrt{d^2 + 4h^2}$ (2) mgh
(3) mgd (4) $\frac{mg}{2h} \sqrt{h^2 + d^2}$

38. A boy sitting on the topmost berth in the compartment of a train, which is just going to stop on a railway station, drops an apple aiming at the open hand of his brother sitting vertically below his hands at a distance of about 2 m. The apple will fall
- precisely on the hand of his brother
 - slightly away from the hand of his brother in the direction of motion of the train
 - slightly away from the hand of his brother in the direction opposite to the direction of motion of the train
 - None of the above

39. A bullet, moving with a speed of 150 ms^{-1} , strikes a wooden plank. After passing through the plank, its speed becomes 125 ms^{-1} . Another bullet of the same mass and size strikes the plank with a speed of 90 ms^{-1} . Its speed after passing through the plank would be
- 25 ms^{-1}
 - 35 ms^{-1}
 - 50 ms^{-1}
 - 70 ms^{-1}

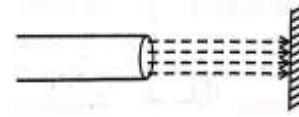
40. A force acts on a 30 g particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is in metres and t is in seconds. The work done during the first 4 s is
- 5.28 J
 - 450 mJ
 - 490 mJ
 - 530 mJ

41. A cart having total mass 3000 kg is moving with constant speed 10 m/sec. Now if sand is leaking from hole with rate 10 kg/sec, then speed of cart after 2 minute will be:



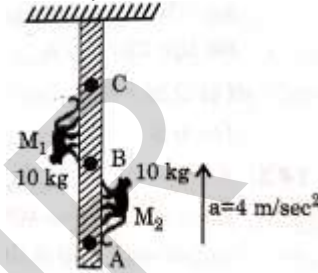
- 10 m/sec
- 16.67 m/sec
- 15 m/sec
- None

42. A water jet having mass per unit length $\lambda = 100 \text{ kg/m}$ and velocity of water coming out is $v = 2 \text{ m/sec}$, force exerted by jet on wall will be:



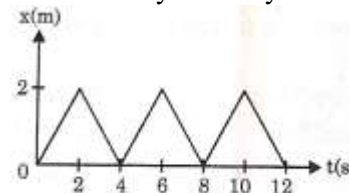
- 200 N
- 400 N
- 100 N
- Can't calculate

43. If monkey M_2 is moving upward with an acceleration $a = 4 \text{ m/sec}^2$ and M_1 is moving with constant velocity. Then T_B and T_C will be:



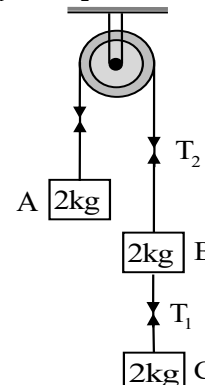
- 100, 240
- 140, 200
- 140, 240
- None

44. Figure shows the position-time ($x-t$) graph of one-dimensional motion of a body of mass 500 g. What is the time interval between two consecutive impulses received by the body?



- 2 s
- 4 s
- 6 s
- 8 s

45. Calculate T_1 and T_2 for the following diagram.



- $\frac{2g}{3}, \frac{4g}{3}$
- $\frac{4g}{3}, \frac{8g}{3}$
- $\frac{6g}{3}, \frac{8g}{3}$
- $\frac{8g}{3}, \frac{4g}{3}$

CHEMISTRY

46. In which of the following dimer empty atomic orbital of central atom of monomer does not involve in hybridisation?
 (1) Ga_2H_6 (2) Al_2Br_6
 (3) Be_2H_6 (4) Cl_2O_6
47. Correct order of ionic size is $\text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$ which can be explained by-
 (1) Nuclear charge (z)
 (2) Principal quantum number (n)
 (3) $e^- - e^-$ interaction in outer orbitals
 (4) None of above because size order is $\text{F}^- = \text{Na}^+ = \text{Mg}^{2+}$
48. Write the period number, group number and block of the element having atomic number 42.
 (1) 5, 5, d (2) 5, 6, d
 (3) 5, 2, d (4) 5, 15, P
49. Which element is related from s-block
 (1) $[\text{Ar}] 3d^5 4s^1$ (2) $[\text{Ar}] 3d^{10} 4s^1$
 (3) $[\text{Kr}] 5s^1$ (4) $[\text{Xe}] 4f^{14} 5d^{10} 6s^2$
50. The boiling point of ICl is nearly 40°C higher than that of Br_2 although the two substances have the same relative molecular mass. This is because:-
 (1) ICl is ionic compound
 (2) I-Cl bond is stronger than Br-Br bond
 (3) ICl is polar covalent molecule while Br_2 is non polar
 (4) IP of Iodine is less than that of Br.
51. Strongest hydrogen bond is shown by:
 (1) Water
 (2) Ammonia
 (3) Hydrogen fluoride
 (4) Hydrogen Sulphide
52. The correct order of volatility is :-
 (1) $\text{NH}_3 < \text{H}_2\text{O}$
 (2) $p\text{-nitrophenol} < o\text{-nitrophenol}$
 (3) $\text{CH}_3\text{OH} > \text{CH}_3 - \text{O} - \text{CH}_3$
 (4) $\text{HF} > \text{HCl}$
53. Which of the following elements cannot exhibit sp^3d hybridisation state?
 (a) C (b) P (c) Cl (d) B
 Correct answer is:-
 (1) a, c (2) a, d (3) b, c (4) b, d
54. The bonds present in N_2O_5 are :-
 (1) Only ionic
 (2) Covalent & coordinate
 (3) Only covalent
 (4) Covalent & ionic
55. The number of antibonding electron pairs in O_2^{2-} molecular ion on the basis of molecular orbital theory is (at. no. O = 8) :
 (1) 2 (2) 3
 (3) 4 (4) 5
56. The bond length in LiF will be :
 (1) less than that of NaF
 (2) equal to that of KF
 (3) more than that of KF
 (4) equal to that of NaF
57. **Assertion :** Ionic compounds do not exhibit stereo isomerism.
Reason : Ionic bonds are non-directional.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is False.
 (4) If both Assertion & Reason are False.
58. The number of elements in 5th and 6th period of periodic table are respectively:
 (1) 8, 18 (2) 18, 18
 (3) 18, 32 (4) 18, 28
59. The IUPAC name of the elements which is placed after DB_{105} in the periodic table, will be:-
 (1) Unnilpentium (2) Unnilium
 (3) Unnilhexium (4) Unnilquadium
60. In sodium atom, the screening is due to:-
 (1) $3s^2, 3p^6$ (2) $2s^1$
 (3) $1s^2, 2s^2 2p^6$ (4) $1s^2, 2s^2$

61. Match the column.

Column-I		Column-II	
Valence electronic configuration		Successive ionisation energies	
(a)	ns^1	(p)	19, 27, 36, 48, 270
(b)	ns^2	(q)	16, 28, 34, 260
(c)	$ns^2 np^1$	(r)	18, 26, 230, 250,
(d)	$ns^2 np^2$	(s)	14, 200, 220, 240,

- (1) (a) s, (b) r, (c) q, (d) p
 (2) (a) r, (b) s, (c) q, (d) p
 (3) (a) s, (b) q, (c) r, (d) p
 (4) (a) s, (b) r, (c) p, (d) q

62. The strength of bonds by $2s - 2s$, $2p - 2p$ and $2p - 2s$ overlapping has the order:-

- (1) $s - s > p - p > s - p$
 (2) $s - s > p - p > s - p$
 (3) $s - s < s - p < p - p$
 (4) $p - p > s - s > p - s$

63. Which of the following ion do not have bond order of 2.5?

- (1) O_2^- (2) O_2^+
 (3) N_2^+ (4) N_2^-

64. Correct order of covalent character of alkaline earth metal chloride is

- (1) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
 (2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
 (3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
 (4) $SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$

65. On the basis of hybridisation when one s & one p orbitals are mixed they are arranged like :-

- (1) Two orbitals mutually at 90° angle
 (2) two orbitals mutually at 180° angle
 (3) Two orbitals mutually at 120° angle
 (4) Two orbitals mutually at 150° angle

66. In which of the following only $p\pi - p\pi$ bond is present :-

- (1) CO_2 (2) SO_3
 (3) SO_2 (4) XeO_3

67. Which bond length in PCl_5 is large:-

- (1) Equatorial (2) Axial
 (3) Both same (4) Can't predict

68. Dative Bond is absent in :-

- (1) H_2SO_4 (2) NO_3
 (3) N_2O (4) H_2CO_3

69. Which of the following is a non-polar molecule?

- (1) CH_3Cl (2) SO_2
 (3) NH_3 (4) SF_6

70. Which of the following elements have electron affinity greater than S?

- (1) O (2) Se
 (3) Te (4) None of these

71. Give the correct letter T for true and F for false for the following statements.

- (I) For Mn, the order of energy is $3s = 3p = 3d$ for shell number 3
 (II) 32 elements are present in fifth period of long form of periodic table.
 (III) For Zr^{2+} ion, $\mu_m = \sqrt{8}BM$
 (IV) General valence shell electronic configuration for f-block elements is $(n-2)f^{1-14}(n-1)d^{10}ns^2$.

- (1) TTFT (2) TTFT
 (3) FFTF (4) FTTF

72. Choose the correct order of ionization energy for the following species.

- (1) $Sc > La > Y$ (2) $Sc > Y \approx La$
 (3) $Sc > Y > La$ (4) $Sc < Y > La$

73. Assuming that elements are formed to complete the seventh period, what would be the atomic number of the alkaline earth metal of the eighth period?

- (1) 113 (2) 120 (3) 119 (4) 106

74. The correct order of atomic/ionic radii is:

- (1) $Co > Ni > Cu > Zn$
 (2) $S^{2-} > Cl^- > O^{2-} > N^{3-}$
 (3) $Sc > Ti > V > Cr$
 (4) None of these

75. The electron gain enthalpies of halogens in $kJ\ mol^{-1}$ are as given below: $F = -332$, $Cl = -349$, $Br = -324$, $I = -295$.

The less negative value for F as compared to that of Cl is due to :

- (1) Strong electron-electron repulsions in the compact 2p-sub shell of F.
 (2) Strong electron-electron repulsions in the bigger 3p-sub shell of Cl
 (3) Smaller electronegativity value of F than Cl
 (4) All of these

76. **Statement-I:** The third period contains only 8 elements and not 18 like 4th period.

Statement-II: In III period filling starts from 3s¹ and complete 3p⁶ whereas in IV period it starts from 4s¹ and complete 4p⁶ after 3d¹⁰

- (1) Both the statements are correct and statement II is the correct explanation of statement I.
- (2) Both the statements are correct but statement II is NOT the correct explanation of statement I.
- (3) Statement I is correct and statement II is incorrect.
- (4) Statement I is incorrect and statement II is correct.

77. **Statement-I:** Formation of Cl⁻ ion is exothermic whereas O²⁻ ion formation is endothermic.

Statement-II: ΔH_{eg_2} of oxygen is endothermic and greater than its exothermic ΔH_{eg_1} value of oxygen.

- (1) Both the statements are correct and statement II is the correct explanation of statement I.
- (2) Both the statements are correct but statement II is NOT the correct explanation of statement I.
- (3) Statement I is correct and statement II is incorrect.
- (4) Statement I is incorrect and statement II is correct.

78. **Statement-I:** The 5th period of periodic table contains 18 elements not 32.

Statement-II: $n = 5$, $\ell = 0, 1, 2, 3$. The order in which the energy of available orbitals 4d, 5s and 5p increases is $5s < 4d < 5p$ and the total number of orbitals available are 9 and thus 18 electrons can be accommodated.

- (1) Both the statements are correct and statement II is the correct explanation of statement I.
- (2) Both the statements are correct but statement II is NOT the correct explanation of statement I.

(3) Statement I is correct and statement II is incorrect.

(4) Statement I is incorrect and statement II is correct.

79. The correct order of the metallic character is:

- (1) B > Al > Mg > K
- (2) Al > Mg > B > K
- (3) Mg > Al > K > B
- (4) K > Mg > Al > B

80. Which of the following pairs show reverse properties on moving along a period from left to right and from top to bottom in a group ?

- (1) Atomic radius and electron gain enthalpy (negative value)
- (2) Nuclear charge and ionisation enthalpy
- (3) Ionisation enthalpy and electron gain enthalpy (negative value)
- (4) None of the above.

81.

Column-I		Column-II	
(A)	ing order	(P)	F < O < S < Se
(B)	ing order of electron affinity	(Q)	O < N < F < Ne
(C)	ing order of atomic size	(R)	Na < Mg < Al < Si
		(S)	O ²⁻ < O ⁻ < O < O ⁺

- (1) A → P, Q ; B → R ; C → S
- (2) A → Q, P ; B → S ; C → S
- (3) A → R, S ; B → P ; C → Q
- (4) A → Q, S ; B → S ; C → P

82. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is

- (1) X < Z < Y
- (2) Y < X < Z
- (3) Z < Y < X
- (4) X < Y < Z

83. **Assertion (A):** Metallic character decreases and non-metallic character increases on moving from left to right in a period.

Reason (R): It is due to increase in ionization enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) (A) is false but (R) is true.
- (2) (A) is true but (R) is false.
- (3) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

84. Match the column

Column-I (Elements)		Column-II (Ionization Enthalpy)	
P	$[\text{Xe}]^{54} 6s^2 5d^{10} 4f^{14}$	A	576 kJ / mole
Q	$[\text{Ar}]^{18} 4s^2 3d^{10}$	B	1280 kJ / mole
R	$[\text{Ar}]^{18} 4s^2 3d^{10} 4p^1$	C	900 kJ / mole
S	$[\text{Ne}]^{10} 3s^2 3p^1$	D	578 kJ / mole

- (1) P – B, Q – C, R – D, S – A
- (2) P – A, Q – C, R – B, S – D
- (3) P – D, Q – A, R – C, S – B
- (4) P – C, Q – D, R – A, S – B

85. In which of the following ionic compounds, ΔH_f is negative only due to lattice energy?

- (i) NaF
- (ii) MgO
- (iii) Li_3N
- (iv) Na_2S

- (1) Only (iv)
- (2) Only (iii), (iv)
- (3) Only (ii), (iii), (iv)
- (4) All of these

86. In which of the following species maximum atom can lie in same plane?

- (1) XeF_2O_2
- (2) PCl_5
- (3) AsH_4^+
- (4) XeF_4

87. Which is the following pairs of species have identical shapes?

- (1) NO_2^+ and NO_2^-
- (2) PCl_5 and BrF_5
- (3) XeF_4 and ICl_4^-
- (4) TeCl_4 and XeO_4

88. The correct order of H — M — H bonds angle is:

- (1) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3 < \text{BiH}_3$
- (2) $\text{AsH}_3 < \text{SbH}_3 < \text{PH}_3 < \text{NH}_3$
- (3) $\text{NH}_3 < \text{PH}_3 < \text{BiH}_3 < \text{SbH}_3$
- (4) $\text{BiH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$

89. Match the matrix

Column-I (pair of molecular orbitals of O_2)	Column-II (Both M.O. have)
(A) $\sigma_{1s}, \sigma_{2s}^*$	(P) Same shape
(B) π_{2py}, π_{2pz}	(Q) Same energy
(C) π_{2pz}^*, π_{2pz}^*	(R) Largest energy difference (out of given pairs)
(D) σ_{1s}, σ_{2s}	(S) Same source of atomic orbitals of two participating atoms
	(T) Pair of electron in each

- (1) A → R, B → PQT, C → S, D → PT
- (2) A → R, B → PRS, C → S, D → PQR
- (3) A → S, B → PTS, C → T, D → PQR
- (4) A → P, B → TQS, C → S, D → PQR

90. In a regular octahedral molecule MX_6 , the number of X—M—X bonds at 180° is -

- (1) 3
- (2) 2
- (3) 6
- (4) 4

BIOLOGY

91. How many of the given function/s can be performed by collenchyma?
Mechanical support, Photosynthesis, Secretion, Translocation of organic foods
(1) One (2) Two
(3) Three (4) Four
92. Stilt roots
(a) Arise from lower nodes of stem
(b) Provide support as they are hanging from branches
(c) Present in maize and sugarcane
(d) Help in gaseous exchange
Choose correct option.
(1) (a) and (c) (2) (b) and (c)
(3) (a) and (d) (4) (b) and (d)
93. In cockroaches, the respiratory system consists of a network of trachea that open outside through small holes that are present on lateral side of the body are called
(1) Tracheoles and are of 9 pairs
(2) Spiracles and are of 10 pairs
(3) Sphincters and are of 8 pairs
(4) Ostia and are of 13 pairs
94. Monocot root differs from dicot root in having-
(1) Polyarch xylem bundles
(2) Large and well-developed pith
(3) Both (1) and (2)
(4) Radial vascular Bundle and exarch xylem
95. Mark incorrect for dicot root.
(1) Pith-less developed or absent
(2) Secondary growth can occur
(3) The cortex consists of several layers of thin-walled parenchyma cells
(4) Hypodermis is present
96. Which feature is found in dicot stem?
(1) Xylem V or Y-shaped
(2) Presence of Bundle sheath around each vascular bundle
(3) Well developed pith as monocot root
(4) Presence of casparian strips in endodermis
97. Give symbol for actinomorphic flower with 6 tepals, gamotepalous.
(1) $\oplus P_{3+3}$ (2) $\%P_{(6)}$
(3) $\oplus P_{(3)} + P_{(3)}$ (4) $\% P_6$
98. The stem modified into flat, green and performing the function of leaves in:
(1) *Asparagus* (2) *Euphorbia*
(3) *Opuntia* (4) Australian acacia
99. Read the following statements:
(i) A sterile stamen is called staminode
(ii) Epipetalous stamen is found in lily and an epiphyllous stamen in brinjal
(iii) China rose possess polyadelphous stamen
(iv) Salvia and mustard possess stamens with filaments of variable length
(v) Pea possess Monadelphous as well as diadelphous stamen
How many of the above statements are correct?
(1) Three (2) Two (3) Four (4) Five
100. Below aestivation is absent in

(1) China rose (2) *Cassia*
(3) Cotton (4) Okra (Bhindi)
101. Add the missing floral organs in the given floral formula of family-fabaceae.
 $\% \begin{matrix} \text{♂} \\ \text{♀} \end{matrix} K_{(5)} - A_{1+(9)\underline{G}_1$
(1) C_{1+2+2} (2) $C_{1+2+(2)}$
(3) C_{1+2+3} (4) C_5
102. Cockroach excretes nitrogenous waste through A in the form of B. Identify A and B respectively.
(1) Malpighian tubule, uric acid
(2) Green gland, urea
(3) Malpighian tubule, ammonia
(4) Antennal gland, uric acid
103. Special venous connection between liver and intestine as well as the kidney and lower parts of body are present in frog. The former and later are called
(1) Hepatic portal system
(2) Renal portal system
(3) Hepatic portal system and renal portal system, respectively
(4) Renal portal system and hepatic portal system respectively

104. Which of the option represents correct identification for the given alphabet.



- (1) A- Spermatheca
(2) B- Collateral glands
(3) C- Genital pouch
(4) D-Gonapophyses
105. **Assertion (A):** The cockroach's legs are jointed structure that facilitates body movement.
Reason (R): The exoskeleton comprises hardened plates called sclerites, which are interconnected by thin and flexible articular membranes.
In the light of the above statement, choose the correct answer from the options given below.
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.
106. Identify the following statements as true(T) or false (F).
A. A single spermatheca is present in the 6th segment, which opens into the genital chamber.
B. In cockroaches, sperms are stored together in seminal vesicles and are glued together in the form of bundles
C. Mosaic vision has more sensitivity and less resolution.
D. Brain in cockroach is located in the head region and represented by suboesophageal ganglion.
- | A | B | C | D |
|-------|---|---|---|
| (1) F | T | T | F |
| (2) T | F | T | F |
| (3) F | T | F | T |
| (4) T | F | F | T |
107. A complete set of the mouthparts of the cockroach consists of:
(1) Labrum and labium only
(2) Labium, labrum and tongue only
(3) Labrum, mandibles, tongue, maxillae and labium
(4) Labrum, maxillae and labium only

108. In male cockroach, mushroom gland is present in
(1) 8th, 9th abdominal segments
(2) 6th, 7th abdominal segments
(3) 4th, 5th abdominal segments
(4) 2nd, 3rd abdominal segments
109. Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth
(1) Pharynx→ Oesophagus→ Gizzard→ Crop → Ileum → Colon→ Rectum.
(2) Pharynx→ Oesophagus→ Gizzard→ Ileum → Crop → Colon→ Rectum
(3) Pharynx→ Oesophagus→ Ileum → Crop → Gizzard→ Colon→ Rectum
(4) Pharynx→ Oesophagus→ Crop→ Gizzard→ Ileum → Colon → Rectum
110. Male frog can be distinguished from female frog by the presence of
(1) Vocal sacs and copulatory pad on the first digit of the forelimb
(2) A neck and tail is absent
(3) The hindlimb ends in the five digits
(4) Eyes are bulged and covered by the nictitating membrane
111. Select the correct route for the passage of sperms in male frogs.
(1) Testes→ Bidder's canal→ Kidney→ Vasa efferentia→ Urinogenital duct→ Cloaca
(2) Testes→ Vasa efferentia→ Kidney→ Seminal Vesicle→ Urinogenital duct→ Cloaca
(3) Testes→ Vasa efferentia→ Bidder's canal→ Ureter→ Cloaca
(4) Testes→ Vasa efferentia→ Kidney → Bidder's canal→ Urinogenital duct→ Cloaca

112. Match the following columns

Column I (Tissues)		Column II (Composition)	
A.	Areolar tissue	(i)	Fat cells
B.	Adipose tissue	(ii)	Osteocytes
C.	Ligament	(iii)	Loose connective tissue
D.	Bone cells	(iv)	Dense regular connective tissue

- (1) A -(iii), B-(i), C-(iv), D-(ii)
(2) A-(i), B-(ii), C-(iii), D-(iv)
(3) A-(iv), B-(iii), C-(ii), D-(i)
(4) A-(ii), B-(i), C-(iv), D-(iii)

113. In frog, peripheral nervous system is divided to
 (1) Brain and spinal cord
 (2) Cranial and spinal nerves
 (3) Sympathetic and parasympathetic
 (4) Forebrain, midbrain and hindbrain

114. Match the following with reference to cockroach and choose the correct option.

Column I (Body parts of cockroach)		Column II (Position in the body)	
A.	Ovary	(i)	6 in number
B.	Abdominal ganglion	(ii)	9 th sternum
C.	Phallomeres	(iii)	2 th –6 th segment
D.	Total Abdominal segments	(iv)	10 segments

- (1) A-(iv), B-(iii), C-(ii), D-(i)
 (2) A-(iii), B-(i), C-(ii), D-(iv)
 (3) A-(iv), B-(ii), C-(i), D-(iii)
 (4) A-(i), B-(ii), C-(iii), D-(iv)

Directions (Que.115-121):- Consider the following statements.

- (A) Statement I is correct, but Statement II is incorrect.
 (B) Statement II is correct, but Statement I is incorrect.
 (C) Both Statement I and Statement II are correct.
 (D) Both Statement I and Statement II are incorrect.

Select the correct option:-

115. **Statement-I** Sieve tube members have sieve plates where they join with other sieve tube members.
Statement-II Sieve plates are necessary to allow conduction between Sieve tube cells.
 (1) A (2) B (3) C (4) D
116. **Statement-I** Medullary ray is radial strip of parenchyma present between vascular bundles
Statement-II: Medullary rays perform radial conduction of water & minerals
 (1) A (2) B (3) C (4) D
117. **Statement-I** Mango fruit is developed by bicarpellary, syncarpous and superior ovary
Statement-II Mesocarp is edible in Mango
 (1) A (2) B (3) C (4) D
118. **Statement-I** Epigynous flowers are seen in rose plants.
Statement-II In brinjal, the ovary is superior.
 (1) A (2) B (3) C (4) D

119. **Statement-I** In cockroach, except foregut entire alimentary canal is lined by cuticle.

Statement-II In cockroach, oesophagus opens into a sac-like structure called crop.

- (1) A (2) B (3) C (4) D

120. **Statement-I** In frogs, Eyes are bulged and covered by nictitating membrane

Statement-II Membranous tympanum of frog receives the sound signals

- (1) A (2) B (3) C (4) D

121. Read the following statements (I-IV).

- I. Cockroaches are brown or black bodied animals that are included in class insecta of phylum arthropoda.
 II. Males bear a pair of short, thread like anal styles which are absent in females.
 III. Heart of cockroach consists of elongated muscular tube lying along mid dorsal line of thorax and abdomen.
 IV. The nymph grows by moulting about 13 times to reach the adult form.

Which of the above statements are correct?

- (1) I, II and III only (2) II, III and IV only
 (3) I and II only (4) I, II, III and IV only

122. Bright colour of petals is due to presence of

- (1) Chloroplast (2) Anthocyanin
 (3) Chromoplast (4) Leucoplast

123. **Statement:** While observing transverse sections of two stems, the anatomical characters were recorded as under:

- (A) Vascular bundles conjoint with Sclerechymatous bundle sheath.
 (B) Vascular bundles conjoint without Sclerechymatous bundle sheath
 (C) Vascular bundles closed.
 (D) Vascular bundles open.

Which of the following combinations are features that will assign one of the materials to a dicot stem?

- (1) A and C (2) B and C
 (3) B and D (4) A and D

124. Read the given statements about blood vascular system of cockroach.

- (A) Circulatory system of cockroach is of closed type
 (B) It contains no blood vessels except aorta of heart.
 (C) The haemolymph is composed of colourless plasma and haemocytes

Which of the statement (s) given above is/are incorrect?

- (1) Only A (2) A, B and C
 (3) A and C only (4) Only B

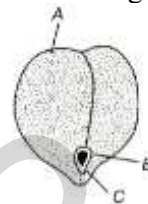
125. Medullary rays are formed by the
- Radially placed parenchymatous cells between vascular bundles
 - Longitudinally placed parenchymatous cells between vascular bundles
 - Laterally placed parenchymatous cells between vascular bundles
 - Obliquely placed parenchymatous cells between vascular bundles
126. Lateral branches with short internodes and each node bearing a rosette of leaves above and a tuft of roots below is found in aquatic plants like.
- Jasmine
 - Eichhornia*
 - Pineapple
 - Strawberry
127. Which of the following is incorrect for the stem?
- The leaf bearing part of the stem is called the bud
 - The shoot is a leafless part of the stem
 - Stems are negatively phototropic
 - Stems are positively geotropic
- Only A
 - Only B
 - Only C and D
 - All A, B, C and D
128. Read the following statements, w.r.t. nervous system of frog and mark the correct option.
- There are ten pairs of cranial nerves arising from the brain.
 - Hind brain consists of pons and medulla oblongata.
 - Spinal cord is not enclosed in vertebral column.
 - Brain is divided into forebrain, midbrain and hind brain.
 - Mid brain is characterised by a pair of olfactory lobes.
 - Paired cerebral hemispheres and unpaired diencephalon are present in frog.
- A, B and C only
 - A, D and E only
 - A, D and F only
 - All of these
129. Match the following with reference to cockroach and choose the correct option

Column-I		Column-II	
A	Phallomere	i	Chain of developing ova
B	Gonopore	ii	Bundles of sperm
C	Spermatophore	iii	Opening of the ejaculatory duct
D	Ovarioles	iv	The external genitalia

- A-(iii), B-(iv), C-(ii), D-(i)
- A-(iv), B-(iii), C-(ii), D-(i)
- A-(iv), B-(ii), C-(iii), D-(i)
- A-(ii), B-(iv), C-(iii), D-(i)

130. Select the correct statement from the ones given below with respect to *Periplaneta americana*.
- Nervous system located dorsally, consists of segmentally arranged ganglia joined by a pair of longitudinal connectives
 - Males bear a pair of short thread like anal styles
 - There are 16 very long Malpighian tubules present at the junctions of midgut and hindgut
 - Grinding of food is carried out by the mouth parts

131. Identify A, B and C in the given diagram



- A-Seed coat, B-Micropyle, C-Hilum
- A-Seed coat, B-Hilum, C-Micropyle
- A-Hilum, B-Seed coat, C-Micropyle
- A-Micropyle, B-Seed coat, C-Hilum

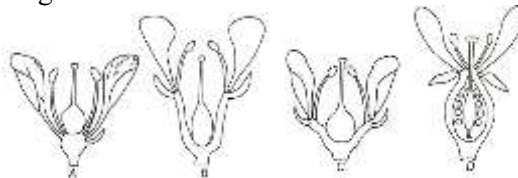
132. Read the following statements w.r.t frog.

- Above the mouth, a pair of nostrils is present.
- On either side of eyes pinna receives sound signals.
- Frogs are omnivores and hence the length of their intestine is reduced.
- The skin of frog is always maintained in a moist condition.
- A neck and a tail are absent in frogs.

Choose the correct option.

- I and II only
- I, IV and V
- II, III and IV
- I, II and V only

133. Identify the position of gynoecium in the given diagrams A to D



- A-Perigynous, B-Perigynous, C-Hypogynous, D-Epigynous
- A-Epigynous, B-Perigynous, C-Hypogynous, D-Perigynous
- A-Hypogynous, B-Perigynous, C-Perigynous, D-Epigynous
- A-Hypogynous, B-Epigynous, C-Perigynous, D-Perigynous

134. Statement I: Testes are adhered to the upper part of the kidneys by a double fold of peritoneum called mesorchium.

Statement II: Vasa efferentia are 50 to 60 in number and enter kidneys on their side.

- (1) Both statement I statement II are correct.
- (2) Statement I is correct but, statement II is incorrect.
- (3) Statement I is incorrect, but statement II are correct.
- (4) Both statement I and Statement II are incorrect.

135. Which cells possess chloroplast and regulate the opening and closing of stomata?

- (1) Cuticle cells (2) Stomatal cells
- (3) Guard cells (4) Subsidiary cells

136. Match the following columns with reference to cockroach and select the correct option.

Column-I		Column-II	
A	Grinding of the food particles	i	Hepatic caeca
B	Secrete gastric juice	ii	10th segment
C	10 pairs	iii	Proventriculus
D	Anal Cerci	iv	Spiracles
		v	Alary muscles

- (1) A-(ii), B-(iii), C-(i), D-(iv)
- (2) A-(iii), B-(i), C-(iv), D-(ii)
- (3) A-(iv), B-(iii), C-(v), D-(ii)
- (4) A-(i), B-(iv), C-(iii), D-(ii)

137. Epidermal tissue system comprises

- (1) Epidermal cells
- (2) Stomata
- (3) Epidermal appendages-the trichomes & hairs
- (4) All of the above

138. Pulvinus is:

- (1) Swollen leaf base
- (2) Modified petiole in Australian acacia
- (3) Small lateral leaf-like structure at leaf base
- (4) Cylindrical stalk that joins the leaf base with the lamina of the leaf

139. Consider the following statements and select the option that correctly fills in the blanks:

- (i) The outermost covering of a seed is called seed coat. It has two coverings, the outer _____ A _____ and inner _____ B _____.
- (ii) _____ C _____ is the scar on the seed coat through which the developing seed is attached to the fruit

A	B	C
(1) Tegmen	Testa	Hilum
(2) Testa	Tegmen	Hilum
(3) Testa	Tegmen	Funicle
(4) Tegmen	Testa	Caruncle

140. Read the following statements-

- A. The development of *Periplaneta* is paurometabolous.
- B. Number of ovarioles in each ovary is 16.
- C. Male cockroach has 3 external genitalia called gonapophysis.
- D. Mouth parts of cockroach are sponging type.

The correct statement are

- (1) A and B (2) B and C
- (3) A and C (4) C and D

141. Collenchyma cells are characterized by:

- A. Presence below the epidermis in most dicotyledonous plants
 - B. Presence as homogenous layer or in patches
 - C. Presence of cells which are much thickened at corners due to the deposition of cellulose, hemicellulose and pectin
 - D. Chloroplast as constant feature
- (1) Only (A) and (C) are correct
 - (2) (A), (C) and (D) are correct
 - (3) (A), (B) and (C) are correct
 - (4) (A), (B), (C) and (D) are correct

142. Vascular system includes...A... bundles, which can be seen in the lateral veins and ...B.... The size of vascular bundles are dependent on the size of...C.... The veins vary in thickness in the reticulate venation of the...D... leaves.

Choose the correct option to replace A to D.

- (1) A-phloem, B-midrib, C-veins, D-dicot
- (2) A-xylem, B-midrib, C-veins, D-dicot
- (3) A-vascular, B-midrib, C-veins, D-dicot
- (4) A-vascular, B-midrib, C-veins, D-monocot

143. Match the Column-I & II and find out correct answer.

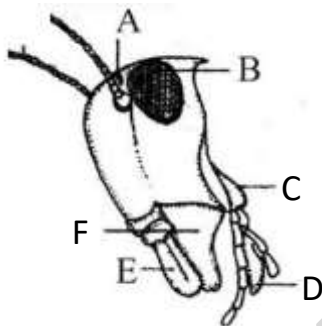
Column-I		Column-II	
a.	Valvate aestivation	(i)	Gulmohur
b.	Imbricate aestivation	(ii)	Pea
c.	Twisted aestivation	(iii)	<i>Calotropis</i>
d.	Vexillary aestivation	(iv)	Cotton

- (1) a-iii, b-i, c-iv, d-ii
- (2) a-iii, b-i, c-ii, d-iv
- (3) a-iv, b-iii, c-i, d-ii
- (4) a-ii, b-iv, c-i, d-iii

144. Read the following statements with respect to reproduction in cockroach and identify the wrong one among them.

- (1) In male, sperms are stored in seminal vesicles and are glued together in the form of bundles called spermatophores.
- (2) Male cockroach has a pair of testes, lying one on each lateral side in the 4th -6th abdominal segments.
- (3) Female cockroach has a pair of spermatheca in 8th segment which opens into the genital chamber.
- (4) On an average female produces 9-10 ootheca, each containing 14-16 eggs.

145. The given figure is related to the head region of cockroach. Identify A to F with the correct combination of options:



- (1) A-Maxilla, B-Compound eye, C-Ocellus, D-Labrum, E-Labium, F - Mandible
- (2) A-Ocellus, B-Compound eye, C-Maxilla, D-Labium E-Labrum, F- Mandible
- (3) A-Ocellus, B-Compound eye, C-Maxilla, D-Labrum E-Labium, E- Mandible
- (4) A-Mandible, B-Compound eye, C-Maxilla, D-Ocellus, E-Labrum, F-Labium

146. Endosperm is a product of double fertilization in angiosperm. Endosperm is not present in the seeds of:

- (1) Castor
- (2) Pea
- (3) Maize
- (4) Coconut

147. Regeneration of damaged growing grass following grazing is largely due to

- (1) Lateral meristem
- (2) Apical meristem
- (3) Intercalary meristem
- (4) Secondary meristem

148. Match the Column-I & II and find out correct answer.

Column-I	Column-II
a. <i>Calotropis</i>	(i) Alternate phyllotaxy
b. Mustard	(ii) Pinnately compound leaf
c. <i>Alstonia</i>	(iii) Opposite phyllotaxy
d. Rachis	(iv) Whorled phyllotaxy

- (1) a-iii, b-i, c-iv, d-ii
- (2) a-iii, b-iv, c-i, d-ii
- (3) a-iv, b-ii, c-i, d-iii
- (4) a-ii, b-iv, c-iii, d-i

149. Match the following column-I & column-II.

Column I (Placentation)	Column II (Examples)
A. Basal	(i) Mustard
B. Axile	(ii) China rose
C. Parietal	(iii) <i>Dianthus</i>
D. Free-central	(iv) Sunflower

- (1) A-(ii), B-(iii), C-(iv), D-(i)
- (2) A- (i), B-(ii), C-(iii), D-(iv)
- (3) A- (iv), B-(ii), C-(i), D-(iii)
- (4) A- (iii), B-(iv), C-(i), D-(ii)

150. Match the following columns.

Column-I (Cells)	Column-II (Functions)
A Bulliform cells	i Regulate opening and closing of stomata
B Guard cells	ii Aerating pores in the bark of plant
C Lenticels	iii Rolling of leaves
D Subsidiary cells	iv Accessory cells

- (1) A-(iii), B-(i), C-(ii), D-(iv)
- (2) A-(i), B-(ii), C-(iii), D-(iv)
- (3) A-(iv), B-(iii), C-(i), D-(ii)
- (4) A-(ii), B-(iv), C-(iii), D-(i)

151. Vexillary aestivation is characteristic of the family

- (1) Fabaceae
- (2) Asteraceae
- (3) Solanaceae
- (4) Brassicaceae

152. Which floral family has (9) + 1 arrangement of anthers in the androecium?

- (1) Malvaceae
- (2) Rutaceae
- (3) Fabaceae
- (4) Caesalpinhiaceae

Direction(Que.153-164): In each of the following questions, a statement of Assertion (A) is given by corresponding statement of Reason (R). Of the statements, mark the correct answers as

- (A) If both A and R are true and R is the correct explanation of A.
 (B) If both A and R are true, but R is not the correct explanation of A.
 (C) If A is true, but R is false.
 (D) If A is false, but R is true.

153. Assertion: In frog, the alimentary canal is short, and length of intestine is reduced.

Reason: Frog is carnivorous in nature.

- (1) A (2) B (3) C (4) D

154. Assertion (A) Root hairs are unicellular and trichomes are usually multicellular.

Reason (R) Root hairs are found in root epidermis and trichomes are found in stem epidermis.

- (1) A (2) B (3) C (4) D

155. Assertion (A) Stems of some plants protect them from browsing animals.

Reason (R) Axillary buds of stems of these plants are modified into thorns.

- (1) A (2) B (3) C (4) D

156. Assertion (A) Fruit is the mature or ripened ovary developed after fertilisation.

Reason (R) Parthenocarpic fruit is formed without fertilisation of the ovary.

- (1) A (2) B (3) C (4) D

157. Assertion (A): Leaf has conjoint, collateral and endarch vascular bundles as monocot stem.

Reason (R): Both leaf and monocot stems have closed vascular bundles.

- (1) A (2) B (3) C (4) D

158. Assertion (A): In frog's vascular system, the composition of lymph is same as that of the blood.

Reason (R) : In frog, lymph lacks RBC's and few proteins.

- (1) A (2) B (3) C (4) D

159. Assertion (A) A simple permanent tissues have cells performing simple functions.

Reason (R): Complex permanent tissues perform specific functions.

- (1) A (2) B (3) C (4) D

160. Assertion (A) Sclereids are found in fruit walls of nuts, pulp of fruits like guava, pear and sapota and seed coats of legumes.

Reason (R) Sclereids are spherical, oval or cylindrical, highly thickened dead cells each with narrow lumen.

- (1) A (2) B (3) C (4) D

161. Assertion(A): Frog shows phenomenon of hibernation.

Reason (R) : Frogs are poikilothermic animals.

- (1) A (2) B (3) C (4) D

162. Assertion: In cockroach, each segment is covered by three hardened plates called sclerites.

Reason: These sclerites are dorsal tergite, ventral sternite, and lateral pleurites.

- (1) A (2) B (3) C (4) D

163. Assertion (A): If the head of a cockroach is cut off, it will remain alive for as long as one week.

Reason (R): The head of cockroach holds only a bit of nervous system.

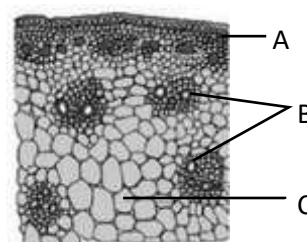
- (1) A (2) B (3) C (4) D

164. Assertion (A): Marginal placentation is observed in marigold plant.

Reason (R): In pea plant, the placenta forms a ridge along the ventral suture of ovary and ovules are borne on this ridge forming two rows.

- (1) A (2) B (3) C (4) D

165. Label correctly A, B and C in the following diagram:-



	A	B	C
(1)	Hypodermis	Vascular bundles	Pith
(2)	Hypodermis	Vascular bundles	Ground tissue
(3)	Epidermis	Vascular bundles	Pith
(4)	Cortex	Vascular bundles	Ground tissue

166. Fill in the blanks by choosing the correct option for A, B and C:

Structure	Example	Function
Thorn	A	Defense
B	Cactus	C

A B C

- (1) *Euphorbia* Leaf spine Stored food
 (2) *Bougainvillea* Leaf spine Defense
 (3) Australian acacia Thorn Defense
 (4) *Opuntia* Thorn Protective
167. In the cymose type of inflorescence:
 A. The main axis terminates in a flower
 B. The flowers are born in acropetal succession
 C. The younger flowers are born at the base of the floral axis

Select the correct statements:

- (1) Only (A) (2) (A) and (B)
 (3) (A) and (C) (4) (A), (B) and (C)

168. Match the following columns and select the correct option:

Column I	Column II
A. Actinomorphic flower	(i) Plum
B. Zygomorphic flower	(ii) China rose
C. Hypogynous flower	(iii) Bilateral symmetry
D. Perigynous flower	(iv) Radial symmetry

- (1) A – (iii), B – (iv), C – (i), D – (ii)
 (2) A – (iii), B – (iv), C – (ii), D – (i)
 (3) A – (iv), B – (iii), C – (ii), D – (i)
 (4) A – (iv), B – (iii), C – (i), D – (ii)

169. Consider the following statements A, B, C and D and select the option for two correct statements:

- A. The ovary is said to be half-inferior in the perigynous flower
 B. In axile placentation, the ovary is one chambered but it becomes two chambered due to formation of false septum called replum
 C. Inferior ovary is found in epigynous flower
 D. When stamens are attached to tepals, they are epiphyllous in brinjal

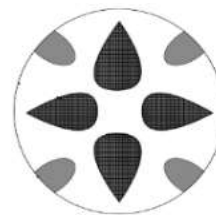
The correct statements are:

- (1) C and D (2) A and B
 (3) A and C (4) B and D

170. Find the incorrectly matched pair:

- (1) Companion cells – Maintain pressure gradient in the sieve tubes
 (2) Sieve tube – Peripheral cytoplasm with large central vacuole and prominent nucleus
 (3) Phloem parenchyma – Stores food materials and other substances like resin, latex and mucilage
 (4) Phloem fibres – Absent in primary phloem but present in secondary phloem

171. Given below is the transverse section of a plant part



Identify the appropriate answer:

- (1) Radial vascular bundles of monocot root
 (2) Depicts tetrarch condition of dicot root
 (3) Depicts conjoint collateral open vascular bundles of dicot stem
 (4) Radial vascular bundle with centrifugal xylem differentiation

172. Match the columns and select the correct option:

Column I	Column II
A. Collenchymatous hypodermis	(i) Root endodermis
B. Starch sheath	(ii) Monocot stem
C. Casparian strip	(iii) Dicot stem
D. Undifferentiated ground tissue	(iv) Endodermis of dicot stem

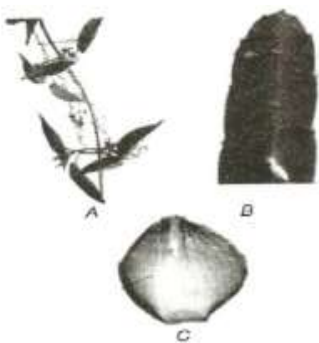
- (1) A – (ii), B – (iv), C – (i), D – (ii)
 (2) A – (ii), B – (iv), C – (i), D – (iii)
 (3) A – (iii), B – (iv), C – (i), D – (ii)
 (4) A – (ii), B – (i), C – (iv), D – (iii)

173. Study carefully the given floral diagram and select the option, which correctly represents the related Floral Formula (FF).



- (1) $\oplus \text{♀} \overbrace{P_{(3+3)} A_{3+3} \underline{G}_{(3)}}$
 (2) $\oplus \text{♀} P_6 A_6 \underline{G}_{(3)}$
 (3) $\oplus \text{♀} P_{5+5} A_{(5)} \underline{G}_{(2)}$
 (4) $\oplus \text{♀} \overbrace{K_{(5)} C_{(5)} A_5 \underline{G}_{(2)}}$

174. Identify the type of leaf modification in the given diagram (A to C).



- (1) A-Support (spines), B-Protection (tendril), C-Storage (fleshy leaves)
- (2) A-Support (tendril), B-Protection (spine), C-Storage (fleshy leaves)
- (3) A-Protection (tendril), B-Support (spine), C-Storage (fleshy leaves)
- (4) A-Protection (spine), B-Support (tendril), C-Storage (fleshy leaves)

175. **Statement-I:** Venus-fly trap are modified leaves.

Statement-II: Parallel venation is found in most monocotyledons.

- (1) Both statement-I and II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Both statement-I and II are incorrect.

176. **Statement-I:** Generally sepals are not green.

Statement-II: Petals are usually green in colour.

- (1) Both statement-I and II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Both statement-I and II are incorrect.

177. Read the following statements:

- (i) A flower may be trimerous, tetramerous or pentamerous when the floral appendages are multiples of 3, 4 or 5
 - (ii) Flower without bract is called ebracteate
 - (iii) The stalk of flower is called Pedicel
 - (iv) Asymmetrical flowers are found in beans.
- How many of the above statements are correct?
- (1) Three
 - (2) Two
 - (3) One
 - (4) Four

178. Match the following Column-I & Column-II.

Column I (Layers of monocot/dicot roots)	Column II (Features)
A. Stele	(i) Innermost layer of cortex
B. Endodermis	(ii) Suberin
C. Casparian strips	(iii) Grasses
D. Dumb-belled shaped guard cells	(iv) All the tissue inner to endodermis

- (1) A – (iv), B – (i), C – (ii), D – (iii)
- (2) A – (iii), B – (iv), C – (ii), D – (i)
- (3) A – (iv), B – (iii), C – (ii), D – (i)
- (4) A – (iv), B – (iii), C – (i), D – (ii)

179. **Statement-I:** Flower is considered as modified shoot.

Statement-II: Here shoot apical meristem changes into floral meristem.

- (1) Both statement-I and II are correct
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Both statement-I and II are incorrect.

180. Which of the following is lateral cambium but not a product of dedifferentiation.

- (1) Interfascicular cambium
- (2) Intrafascicular cambium
- (3) Cork cambium
- (4) All of the above

Answer Key

and

Solution

ANSWER KEY

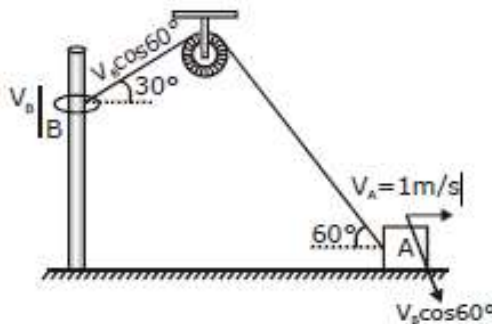
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Ans.	4	4	3	3	3	2	2	4	3	4	1	4	1	3	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	3	3	2	2	1	3	4	2	2	2	2	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	2	1	2	4	2	1	2	2	1	2	2	3	1	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	1	2	3	3	3	2	2	2	3	1	1	3	3	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	3	1	3	2	1	2	4	4	4	3	3	2	3	1
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	1	1	1	4	1	4	4	2	1	3	4	3	4	1	1
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	1	2	3	4	3	3	3	2	2	2	1	3	1	3
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	2	3	2	4	1	4	1	2	2	3	3	2	2	2	3
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	4	2	3	1	1	2	4	3	2	2	2	2	3	2	3
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	2	4	1	2	3	3	3	1	3	2	2	3	1	3	1
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	1	3	1	2	1	2	2	4	2	2	1	4	1	4	2
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	2	3	3	3	2	2	3	1	2	1	4	1	1	1	2

PHYSICS

1. (4)

Sol. Minimum speed at the highest point of vertical circular path $V = \sqrt{gR}$

2. (4)

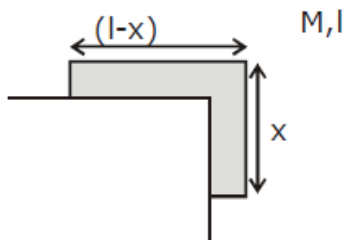


Sol.

$$V_A \cos 60^\circ = V_B \cos 60^\circ \Rightarrow V_A = V_B = 1 \text{ m/s}$$

3. (3)

Sol.



$$\text{Mass of length } x = \frac{M}{l}x$$

$$\text{mass of } (1-x) \text{ length} = \frac{M}{l}(1-x)$$

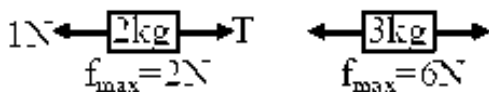
$\therefore$ at equilibrium

$$\frac{M}{l}xg = \mu \frac{M}{l}(1-x)g$$

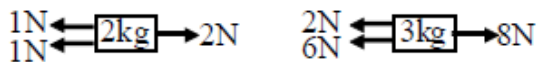
$$X = \frac{\mu l}{1+\mu}$$

4. (3)

sol. FBD



Net force without friction on system is '7N' in right side so first maximum friction will come on 3 kg block



So $f_1 = 1 \text{ N}$, $f_2 = 6 \text{ N}$, $T = 2 \text{ N}$

5. (3)

Sol. Resultant downward force along the incline
 $= mg(\sin\theta - \mu\cos\theta)$

Normal reaction $= mg\cos\theta$

Given : $mg\cos\theta = 2mg(\sin\theta - \mu\cos\theta)$

By solving $\theta = 45^\circ$

6. (2)

Sol. $\mu = \tan(\text{angle of repose}) = \tan 60^\circ = 1.732$

7. (2)

Sol. Fraction of length of the chain hanging from the table

$$\frac{1}{n} = \frac{60\text{cm}}{200\text{cm}} = \frac{3}{10} \Rightarrow n = \frac{10}{3}$$

L/n

Work done in pulling the chain on the table

$$W = \frac{mgL}{2n^2}$$

$$= \frac{4 \times 10 \times 2}{2 \times (10/3)^2} = 3.6 \text{ J}$$

8. (4)

Sol. $U = \frac{1}{2}kx^2$ if x becomes 5 times then energy will become 25 times i.e. $4 \times 25 = 100 \text{ J s}$

9. (3)

10. (4)

$$\begin{aligned} \text{Sol. } W &= \int_0^5 F dx = \int_0^5 (7 - 2x + 3x^2) dx = \left[7x - x^2 + x^3 \right]_0^5 \\ &= 35 - 25 + 125 = 135 \text{ J} \end{aligned}$$

11. (1)

$$\text{Sol. } Mg - N = \frac{mv^2}{R}, N = 0$$

$$v = \sqrt{Rg}$$

12. (4)

Sol. Area between curve and displacement axis

$$= \frac{1}{2} \times (12 + 4) \times 10 = 80 \text{ J}$$

In this time body acquire kinetic energy

$$= \frac{1}{2} mv^2 \text{ by the law of conservation of energy}$$

$$\frac{1}{2} mv^2 = 80 \text{ J}$$

$$\Rightarrow \frac{1}{2} \times 0.1 \times v^2 = 80$$

$$\Rightarrow v^2 = 1600$$

$$\Rightarrow v = 40 \text{ m/s}$$

13. (1)

Sol. Power = $\frac{\text{work done}}{\text{time}} = \frac{\text{Pressure} \times \text{Change in volume}}{\text{time}}$

$$= \frac{20000 \times 1 \times 10^{-6}}{1}$$

$$= 2 \times 10^{-2} = 0.02 \text{ W}$$

14. (3)



Sol.

Stable equilibrium: A body when displaced from equilibrium, it experiences force/torque in direction opposite to its displacement.

Unstable equilibrium: When body is displaced, it doesn't experience an opposite force, in order to restore equilibrium. Thus we see that when we displace the ball it doesn't tend to come back to its equilibrium position.

15. (3)

Sol. In string ABCD from first format of constraint

$$V_D = 10 \text{ m/s} \uparrow$$

$$\text{Now, } V_D = \frac{V_H + V_E}{2}$$

$V_H = 10 \text{ m/s} \downarrow$ if upward direction is taken to be positive

$$+10 = \frac{-10 + V_E}{2} \Rightarrow V_E = 30 \text{ m/s} -$$

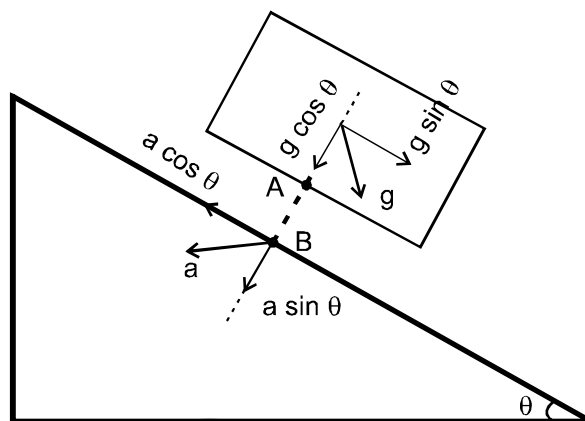
$$\text{Now } \frac{V_F + V_G}{2} = V_E \Rightarrow 30 = \frac{-10 + V_G}{2}$$

$$60 + 10 = V_G$$

$$V_G = 70 \text{ m/s} \uparrow$$

16. (3)

Sol.



acceleration of point A and B must be some along the line $\perp$ to the surface

$$\Rightarrow a \sin \theta = g \cos \theta$$

$$a = g \cot \theta$$

17. (1)

Sol. When F is less than $\mu_s mg$ then tension in the string is zero.

When $\mu_s mg \leq F < \mu_s^2 mg$ then friction on block B is static.

If F is further increased friction on block B is kinetic.

18. (3)

Sol. Here, work done by a car,

$$\text{i.e. } W_g = mgh = 1400 \times 10 \times 10$$

According to work-energy

Theorem, we get

$$W_g + W_f = \frac{1}{2} mv^2$$

$$1400 \times 10 \times 10 + W_f$$

$$= \frac{1}{2} \times 1400 \times 15 \times 15$$

$$\Rightarrow W_f = 17.5 \text{ kJ}$$

19. (3)

Sol. Displacement of ball in t^{th} second,

$$s = gt - \frac{1}{2} g$$

$$\left[\because \frac{1}{2} g(2t-1) \right]$$

$$= g \left(t - \frac{1}{2} \right)$$

$$\Rightarrow s \propto \left(t - \frac{1}{2}\right)$$

Or $s_1 : s_2 : s_3$

$$= \left(1 - \frac{1}{2}\right) : \left(2 - \frac{1}{2}\right) : \left(3 - \frac{1}{2}\right) = 1 : 3 : 5$$

Now, $W = mgs$ ($\because W = Fs$)

$$W \propto S$$

$$\therefore W_1 : W_2 : W_3 = 1 : 3 : 5$$

20. (2)

Sol. $\frac{K_1}{K_2} = \left(\frac{v_1}{v_2}\right)^2$

$$\Rightarrow \frac{1}{2} = \frac{v_1^2}{(v_1 + 2)^2} \quad (\because \zeta_1)$$

$$\Rightarrow v_1^2 + 4v_1 + 4 = 2v_1^2$$

$$\Rightarrow v_1^2 - 4v_1 - 4 = 0$$

$$v_1 = \frac{4 \pm \sqrt{16 + 16}}{2}$$

$$= 2(\sqrt{2} + 1)\text{ms}^{-1} \text{ or } 2(1 - \sqrt{2})\text{ms}^{-1}$$

21. (2)

Sol. When 1 will hit 2, under elastic collision. 1 will come to rest and 2 will gain speed V as balls are identical then same collision occur between 2 and 3 and finally 1 and 2 will come to rest and 3 will move at speed V .

22. (1)

Sol. Mass of neutron, $m_1 = 1$ unit
Mass of nucleus, $m_2 = A$ units

Here, $u_1 = v$ and $u_2 = 0$

Therefore, velocity of neutron after collision,

$$v = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2$$

$$v_1 = \frac{1 - A}{1 + A} v$$

Kinetic energy of neutron after collision,

$$k_2 = \frac{1}{2}(1) \left[\frac{1 - A}{1 + A} \right]^2 v^2 \quad \dots\dots(1)$$

Kinetic energy of neutron before collision.

$$k_2 = \frac{1}{2}(1)v^2 \quad \dots\dots(2)$$

$$\frac{k_2}{k_1} = \left(\frac{1 - A}{1 + A} \right)^2 = \left(\frac{A - 1}{A + 1} \right)^2$$

23. (3)

Sol. Let masses of fragments are $\frac{2m}{5}$, $\frac{2m}{5}$ and $\frac{m}{5}$.

Let speed of third fragment is $u \hat{r}$, if first two fragments fly along positive x-axis and y-axis respectively at speed v . Using conservation of linear momentum,

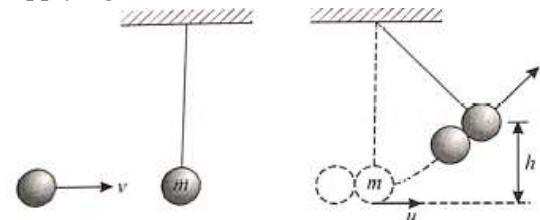
$$m \times 0 = \frac{2m}{5} v \hat{i} + \frac{2m}{5} v \hat{j} + \frac{m}{5} u \hat{r}$$

$$u \hat{r} = -(2v) \hat{i} - (2v) \hat{j}$$

$$|\vec{r}| = \sqrt{(-2v)^2 + (-2v)^2} = \sqrt{4v^2 + 4v^2} = 2\sqrt{2}v$$

24. (4)

Sol. Applying conservation of linear momentum,



$$mv = 2mu$$

$$\Rightarrow u = \frac{v}{2}$$

using work-energy theorem

$$\frac{1}{2}(2m)u^2 - (2m)gh = 0$$

$$\Rightarrow h = \frac{u^2}{2g} = \frac{v^2}{8g}$$

25. (2)

Sol. Metal ball : $\Delta P = mv$

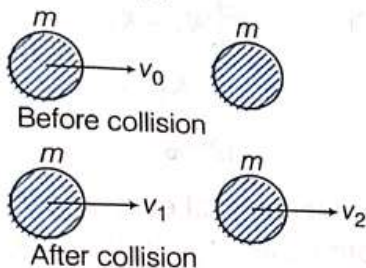
Rubber ball : $\Delta P = 2mv$

$\Rightarrow$ Change in momentum is more for rubber ball.

26. (2)

Sol. $\frac{1}{2}mv_2^2 + \frac{1}{2}mv_1^2 = \frac{150}{100} \times \frac{1}{2}mv_0^2$

$$v_2^2 + v_1^2 = \frac{3}{2}v_0^2 \quad \dots\dots\dots(i)$$



Conservation of momentum gives,

$$mv_0 = mv_1 + mv_2$$

$$v_0 = v_2 + v_1 \quad \dots\dots\dots(ii)$$

Squaring both sides,

$$v_0^2 = v_2^2 + v_1^2 + 2v_1v_2 \quad \dots\dots\dots(iii)$$

From Eqs. (i) and (iii), we have

$$2v_1v_2 = \frac{-v_0^2}{2}$$

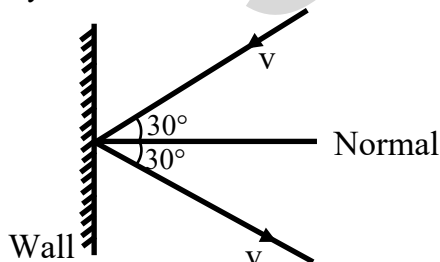
$$\therefore (v_1 - v_2)^2 = (v_1 + v_2)^2 - 4v_1v_2$$

$$= 2v_0^2$$

or $v_{rel} = \sqrt{2}v_0$

27. (2)

Sol. Linear momentum remains conserved in elastic collision as well as in inelastic collision of system but kinetic energy remains conserved only in an elastic collision.



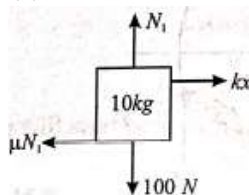
Given that the speed of the molecule of gas is same before and after the collision, therefore its kinetic energy is conserved. Hence, the collision is elastic.

28. (2)

Sol. In equilibrium of a body, vector sum of all the forces acting on bodies is zero so at equilibrium, acceleration of body must always be zero.

29. (1)

Sol.

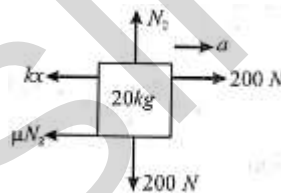


$$N_1 = 100$$

$$kx - \mu N_1 = 10 \times 12$$

$$kx - 10 = 120$$

$$kx = 130 \quad \dots\dots\dots(i)$$



$$N_2 = 200$$

$$200 - kx - \mu N_2 = 20a$$

$$200 - 130 - (0.1 \times 200) = 20a$$

$$70 - 20 = 20a$$

$$a = \frac{50}{20} = 2.5 \text{ m/s}^2$$

30. (2)

Sol. According to question, 7th coin is under the weight of 5 coins above it.

Hence, reaction of the 7th coin on the 8th coin

= force on the 7th coin due to 8th coin

= weight of 5 coins

$$= 5mg = 5 \times 20 \times 10^{-3} \times 10 = 1\text{N}$$

$$\left[\because m = 20 \times 10^{-3} \text{ kg} \right]$$

31. (1)

Sol. When the train accelerates, the box tends to remain at rest (due to inertia). To make the box accelerate with the train, a frictional force acts on it in the direction of the train's motion.

32. (2)

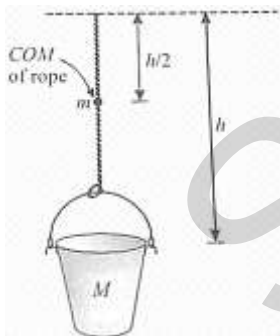
Sol. $F_x = -\frac{\partial U}{\partial x} = -\frac{\partial}{\partial x} \left(\frac{10xy}{z} \right)$
 $= -\frac{10y}{z}$
 $F_y = -\frac{\partial U}{\partial y} = -\frac{\partial}{\partial y} \left(\frac{10xy}{z} \right) = -\frac{10x}{z}$
 $F_z = -\frac{\partial U}{\partial z} = -\frac{\partial}{\partial z} \left(\frac{10xy}{z} \right) = \frac{10xy}{z^2}$
 $\therefore \mathbf{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$
 $= -\left(\frac{10y}{z} \right) \hat{i} - \left(\frac{10x}{z} \right) \hat{j} + \left(\frac{10xy}{z^2} \right) \hat{k}$

33. (1)

Sol. For a body driven by a constant force, power $P = \vec{F} \cdot \vec{v}$ and since direction of $\vec{F}$ and $\vec{v}$ is same so $P = Fv$ and also we can use $v = at$ so $P = F(at)$ which is a linear function.

34. (2)

Sol. $W = mg \frac{h}{2} + Mgh$
 $W = \left(M + \frac{m}{2} \right) gh$



35. (4)

Sol. Given mass and initial velocity
 $m = 0.01 \text{ kg}$
 $\vec{v} = (6\hat{k}) \text{ m/s}$
 $|\vec{v}| = \sqrt{(6)^2 + (16)^2}$
 $|\vec{v}| = \sqrt{36 + 256}$
 $|\vec{v}| = \sqrt{292}$
 $|\vec{v}| = 17.08 \text{ m/s}$
 $\vec{v} = 17.08 \hat{j}$
 $|\vec{v}| = \sqrt{(20)^2 + (17.08)^2}$

$|\vec{v}| = \sqrt{(20)^2 + (17.08)^2}$

$W = \vec{F} \cdot \vec{v}$

$W = m \vec{v} \cdot \vec{v}$

$W = m \left(\frac{v^2 - u^2}{2} \right)$

$W = 0.01 \times \left[\frac{464 - 272}{2} \right]$

$W = 0.01 \times 96$

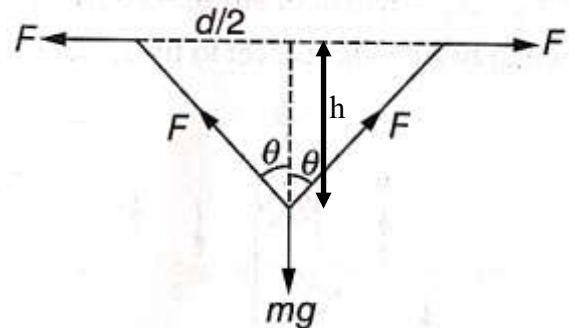
$W = 0.96 \text{ J}$

36. (2)

Sol. $v_i = 20 \text{ m/s}$, $v_f = 0$
 $W_{\text{net}} = \Delta KE = v_f - v_i$
 $= \frac{1}{2} m (v_f^2 - v_i^2)$
 $= \frac{1}{2} \times (0^2 - 20^2)$
 $= -400 \text{ J}$

37. (1)

Sol. $2F \cos \theta = mg$
 $\Rightarrow F = \frac{mg}{2 \cos \theta}$



$\Rightarrow F = \frac{mg \sqrt{h^2 + \frac{d^2}{4}}}{2h} = \frac{mg}{4h} \sqrt{d^2 + 4h^2}$

38. (2)

Sol. Horizontal velocity of apple will remain same but due to retardation of train, velocity of train and hence velocity of boy w.r.t. ground decreases, so apple falls away from the hand of boy in the direction of motion of the train.

39. (2)

Sol. By Work-Energy Theorem

$$\frac{1}{2}m(v^2 - u^2) = Fs \quad \dots(i)$$

For the second case

$$\frac{1}{2}m(v'^2 - u'^2) = Fs \quad \dots(ii)$$

where F is the retarding force offered by the plank to the bullet. Equating (i) and (ii), we get
 $v^2 - u^2 = v'^2 - u'^2$

$$\Rightarrow (125)^2 - (150)^2 = v'^2 - (90)^2$$

$$\Rightarrow v'^2 = (35)^2$$

$$\Rightarrow v' = 35 \text{ ms}^{-1}$$

40. (1)

Sol. $v = \frac{dx}{dt} = 3 - 8t + 3t^2$

$$\Rightarrow u = v|_{t=0} = 3 - 8t + 3t^2 = 3 \text{ ms}^{-1} \text{ and}$$

$$v = v|_{t=4} = 3 - 8t + 3t^2 = 19 \text{ ms}^{-1}$$

$$\Rightarrow W = \frac{1}{2}m(v^2 - u^2)$$

[From work Energy Theorem]

$$\Rightarrow W = \frac{1}{2} \times 0.03 \times (19^2 - 3^2) = 5.28 \text{ J}$$

41. (2)

Sol. $F_{\text{ext}} = 0$ [on system]

$$\frac{mdv}{dt} + \frac{vdm}{dt} = 0$$

$$\frac{mdv}{dt} = -\frac{vdm}{dt}$$

$$(M_0 - 10t) \frac{dv}{dt} = -v(-10) = 10v$$

$$\int_{10}^v \frac{dv}{v} = \int_0^t \frac{10}{M_0 - 10t} dt$$

$$[\ell \quad \quad \ell]$$

$$\ell \ln v/10 = -\ell \ln \frac{M_0 - 10t}{M_0}$$

$$\frac{v}{10} = \frac{M_0}{M_0 - 10t}$$

$$v = \frac{10 \times 3000}{3000 - 120 \times 10} = \frac{10 \times 3000}{1800} = 16.67 \text{ m/s}$$

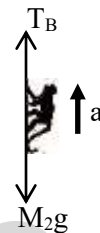
42. (2)

Sol. In every one second, water travel = 2m.

In 2m length, mass of water = $100 \times 2 = 200 \text{ kg}$.
 Force exerted by jet on wall = change in momentum per second
 $= 200 \times 2 = 400 \text{ N}$.

43. (3)

Sol. Free body diagram of M_2

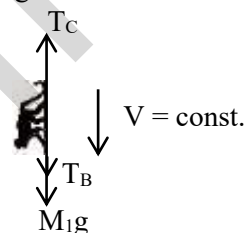


$$T_B - M_2g = M_2a$$

$$T_B = 10 \times 10 + 10 \times 4$$

$$T_B = 140 \text{ N}$$

Free body diagram of M_1



$$T_C = T_B + M_1g$$

$$T_C = 140 + 10 \times 10$$

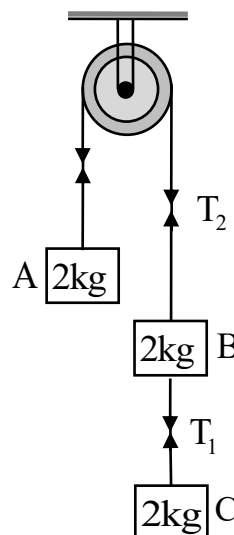
$$= 240 \text{ N}$$

44. (1)

Sol. Velocity change in every 2 second. Therefore impulse received by the body in 2 second.

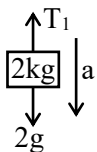
45. (2)

Sol.



$$a = \frac{\text{net pulling force}}{\text{total mass}} = \frac{4g - 2g}{6} = \frac{g}{3} \text{ m/s}^2$$

Now free body diagram of C

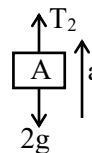


$$2g - T_1 = 2a$$

$$T_1 = 2g - 2a \quad [\because \quad]$$

$$T_1 = 2g - 2 \times g/3 = \frac{4g}{3} \text{ N}$$

Now free body diagram of A



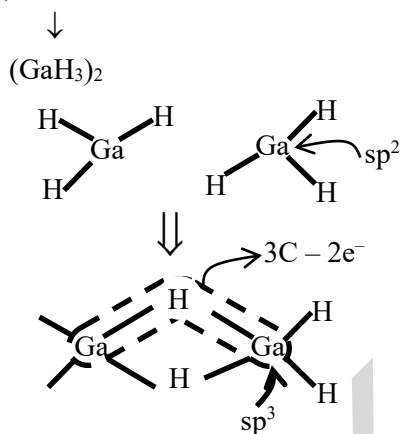
$$T_2 - 2g = 2a$$

$$T_2 = 2 \times g/3 + 2g = \frac{8g}{3} \text{ N}$$

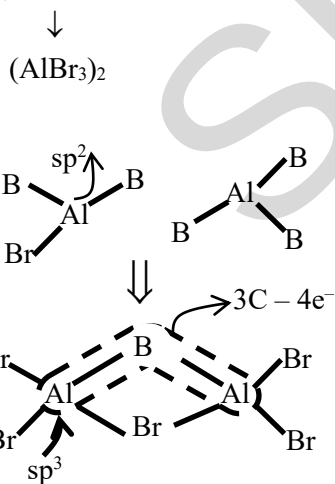
CHEMISTRY

46. (4)

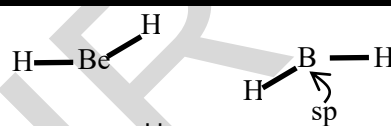
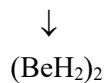
Sol: (1) Ga_2H_6



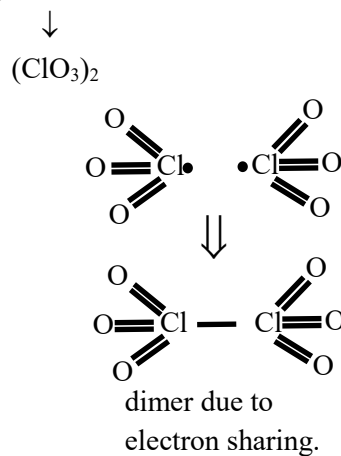
(2) Al_2Br_6



(3) Be_2H_4



(4) Cl_2O_6



47. (1)

Sol:-

	F^-	Na^+	Mg^{2+}
z	9	11	12
e^-	10	10	10

$$Z) = z - s$$

$$Z) \text{ size } ^-$$

48. (2)

Sol. Mo has atomic number 42 with electronic configuration $[\text{Kr}] 5s^1 4d^5$. It belongs to group 6 in period 5 and block d.

49. (3)

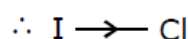
Sol. s-block elements does not have electron in $(n-2)f$; $(n-1)d$ & np subshell.

50. (3)

Sol. ICl bond is polar in nature.

Br_2 is non-polar in nature.

As Cl in ICl is more electronegative.



And in Br_2 both have same electronegativity.



51. (3)

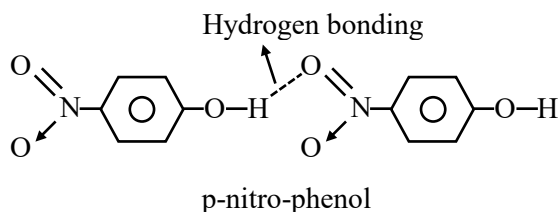
Sol. Hydrogen Bond strength $\propto \Delta\text{EN}$

In HF, ΔEN is maximum so strength of hydrogen bond is maximum.

52. (2)

Sol. Vander waal force $\propto \frac{1}{\text{volatility}}$

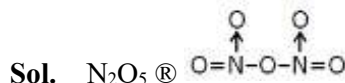
in p-nitro phenol hydrogen bond is present between two molecules; but in o-nitro phenol vander waal bond is present between two molecules & hydrogen bond strength is more stronger than vander waal bond. So, o-nitro phenol have higher volatility.



53. (2)

Sol. Here, carbon and boron cannot exhibit sp^3d hybridisation state as they don't have d-orbital.

54. (2)



Covalent and Coordinate bonds are present.

55. (3)

Sol. The electronic configuration of O_2^{2-} can be written as:

$$s1s^2, s^*1s^2, s2s^2, s^*2s^2, s2p_z^2, p2p_x^2 = p2p_y^2, p^*2p_x^2 = p^*2p_y^2$$

Hence no. of antibonding electrons

$$= 2 + 2 + 2 + 2 = 8$$

$$\text{Antibonding electron pairs} = \frac{8}{2} = 4$$

56. (1)

Sol. LiF is more covalent than NaF and KF.

As covalent nature increases, bond length decreases. Bond length in LiF is less than that of NaF and KF. Size of Na^+ and K^+ is greater than Li^+ .

Bond length is given by distance between nucleus of two atoms.

57. (1)

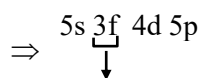
Sol. Ionic compounds do not exhibit stereoisomerism because ionic bonds are non-directional.

58. (3)

Sol. We know that for $n = 5$

The orbitals available for filling are $ns (n-2)f (n-1)d np$

$$\therefore \text{ for } n = 5$$



does not exist

$\therefore$ orbitals available for filling $\Rightarrow 5s \ 4d \ 5p$

maximum e^- that can be filled $\Rightarrow 2 + 10 + 6 = 18$ in these orbitals, so 18 elements are present in 5th period.

Similarly, for 6th period orbitals to be filled.

$$\Rightarrow 6s \ 4f \ 5d \ 6p$$

$$\text{maximum } e^- \text{ filled to be } \Rightarrow 2 + 14 + 10 + 6 \Rightarrow 32$$

$$\Rightarrow 32 \text{ elements.}$$

59. (3)

Sol. After Db 105 $\Rightarrow Z = 106$
Unnilhexium

60. (3)

Sol. The outermost e^- in Na $3s^1$ therefore q is due to $1s^2, 2s^2, 2p^6$.

61. (1)

Sol. The total number of e^- are removed before the highest jump in ionisation energy are the valence e^-

62. (3)

Sol. Same value of n, Bond strength $\propto$ Directional Character
 $2s-2s < 2s-2p < 2p-2p$

63. (1)

Sol. By MOT, Bond Order

$$O_2^- \rightarrow 1.5$$

$$O_2^+ \rightarrow 2.5$$

$$N_2^+ \rightarrow 2.5$$

$$N_2^- \rightarrow 2.5$$

64. (3)

Sol. Polarisation $\propto \frac{1}{\text{Size of cation}}$

Size of cation increases, polarisation decreases, covalent character decreases.

Greatest polarising power of Be^{2+} , shows its maximum covalent character.

65. (2)

Sol. The two orbitals mutually arrange at 180° (sp hybridisation and minimum repulsion).

66. (1)

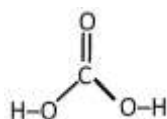
Sol. CO_2 has $p\pi-p\pi$ present (both C & O only contain 2p orbitals.)

67. (2)

Sol. PCl_5 has greater axial Bond length than equatorial (more % p-character)

68. (4)

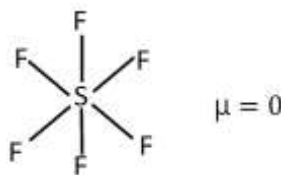
Sol.



H_2CO_3 dative bond is absent

69. (4)

Sol.



SF_6 is a symmetrical molecule (surrounding atom is same, lone pair is absent)

70. (4)

Sol. Electron affinity of S is most in group 16 because of greater size and less repulsion for incoming electron in sulphur.

71. (3)

Sol. (I) Lower the $n + \ell$ value, lower the energy
thus, $3s < 3p < 3d$

(II) 32 elements are present in 6th & 7th period of periodic table.

(III) $Zr^{2+} = [kr]4d^2$

$$\text{So } \mu = \sqrt{n(n+2)} = \sqrt{8} \text{ B.M.}$$

(IV) General electronic configuration of f-block elements is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.

72. (3)

Sol. IE of group 3 is decreasing down the group
 $Sc > Y > La$ (No Contractions)

72. (2)

Sol. Total 118 elements are present in periodic table, so atomic number of alkaline earth metal in eight periodic is 120.

74. (3)

Sol. In 3d series atomic radius is in decreasing order from Sc to Cr.

75. (1)

Sol.

due to smaller in size there is strong repulsion between electrons.

76. (1)

Sol. 3rd period has only 8 elements due to filling form $3s^1$ to $3s^2 3p^6$.

77. (1)

Sol. Formation of Cl^{-1} ion is exothermic whereas formation of O^{-2} is endothermic because EA_2 of a negative ion is more endothermic than EA_1 exothermic

78. (1)

Sol.

79. (4)

Sol.

80. (1)

Sol.

81. (4)

Sol. (A) Increasing order of I.E.

(Q) $\text{O} < \text{N} < \text{F} < \text{Ne}$

(S) $\text{O}^{-2} < \text{O}^{-1} < \text{O} < \text{O}^{+1}$

(B) Increasing order of EA

(S) $\text{O}^{-2} < \text{O}^{-1} < \text{O} < \text{O}^{+1}$

(C) Increasing order of atomic size

(P) $\text{F} < \text{O} < \text{S} < \text{Se}$

82. (4)

Sol.

83. (2)

Sol. From left to right in periodic table :-
Metallic character decreases
Non-metallic character increases
It is due to increase in ionization enthalpy and increase in electron gain enthalpy.

84. (1)

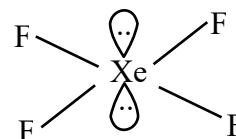
Sol.

85. (3)

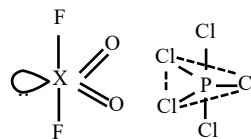
Sol. Formation of oxide, nitride sulphide is endothermic and fluoride is exothermic

86. (4)

Sol.



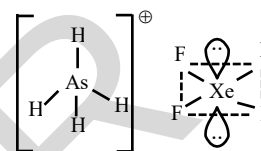
All F and Xe located same plane.



Maximum atoms that →
can lie in a plane

3

4



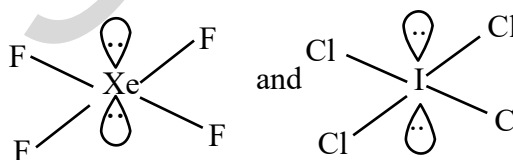
Maximum atoms that →
can lie in a plane

3

5

87. (3)

Sol.



XeF_4 , ICl_4^- Both are square planar in shape

88. (4)

Sol. Due to CA size increase $\text{BL} \uparrow$

$\text{BP} - \text{BP Repulsion} \downarrow$ $\text{BA} \downarrow$

89.

Ans. (1)

90. (1)

Sol.

